

Project Design Phase Proposed Solution Template

Date	1 June 2026
Team ID	SWTID-2026-8236
Project Name	Credit Card Approval Prediction System
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No	Parameter	Description
1	Problem Statement (Problem to be Solved)	Banks receive thousands of credit card applications every day. Manual verification of applicants is time-consuming, expensive, and may lead to inconsistent decisions. The objective of this project is to develop a Machine Learning-based Credit Card Approval Prediction System that accurately predicts whether a credit card application should be approved or rejected based on an applicant's financial and personal details.
2	Idea / Solution Description	The proposed solution is a web-based application developed using Python, Flask, Scikit-learn, HTML, CSS, and JavaScript . The system collects applicant information such as age, income, debt, employment status, credit score, previous default history, and other financial details. After preprocessing the data, a trained Machine Learning model (Random Forest) predicts whether the applicant is eligible for credit card approval. The application also displays approval and rejection probabilities instantly.
3	Novelty / Uniqueness	The system combines machine learning with a simple and user-friendly web interface to provide instant credit card approval predictions. It compares multiple algorithms such as Logistic Regression, Decision Tree, Random Forest, and XGBoost to select the best-performing model. The application provides prediction probabilities instead of only Yes/No results, making the decision process more transparent and reliable.
4	Social Impact / Customer Satisfaction	This solution helps banks reduce manual workload and improve decision-making speed. Customers receive quick eligibility predictions without waiting for lengthy

		verification processes. The system promotes fairness, consistency, and transparency in credit approval while reducing operational costs and improving customer satisfaction.
5	Business Model (Revenue Model)	The application can be offered as a Software-as-a-Service (SaaS) platform to banks, NBFCs, and financial institutions. Revenue can be generated through annual software licensing, subscription plans, API integration for fintech companies, cloud-based deployment services, and custom model development for organizations.
6	Scalability of the Solution	The solution is designed using scalable technologies such as Python, Flask, and Machine Learning, making it suitable for cloud deployment. It can handle thousands of prediction requests simultaneously and can be extended by integrating real-time banking data, credit bureau APIs, fraud detection modules, loan approval systems, and mobile applications.